Edge-Optimized Pure ViT Architectures

A Comprehensive Analysis for Efficient Engineering Workflows

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In the pursuit of **efficient engineering**—particularly for low-resource field deployments requiring real-time continuous analysis, such as automated crop health and soil monitoring systems—pure Vision Transformers (ViTs) offer powerful spatial-temporal reasoning without the computational bloat of traditional CNNs. This document compiles the state-of-the-art open-source ViT models that successfully bypass the quadratic memory explosion of space-time attention, enabling seamless deployment on constrained edge hardware or lightweight Colab environments.

The Pure ViT Edge Leaderboard

MODEL	CREATOR / LAB	CORE EFFICIENCY INNOVATION	PARAMS	COMPUTE	KINETICS-400
Video Swin-T	Microsoft	3D Shifted Window Attention	28M	~42 GFLOPs	81.1%
EVL (ViT-B)	OpenGVLab	Frozen Spatial ViT + Tiny Decoder	~10M	~45 GFLOPs	80.5%
XViT	Samsung AI	Linear Complexity Attention	~40M	Varies	80.2%
TokenLearner (ViT-S)	Google	Dynamic Token Extraction	~22M	~45 GFLOPs	~80.0%
Hiera-Tiny	Meta	Vanilla ViT / MAE Pre-training	28M	~40 GFLOPs	~79.5%
SVT-Small	Duke Uni.	Sparse Motion-driven Attention	24M	~42 GFLOPs	~79.3%
ViViT (Factorized)	Google	Decoupled Space-Time Encoders	27M	~55 GFLOPs	~79.2%
	Tencent		~22M		79.0%

MODEL	CREATOR / LAB	CORE EFFICIENCY INNOVATION	PARAMS	COMPUTE	KINETICS-400
VideoMAE (ViT-S)		Vanilla ViT with 90% Masking		~50 GFLOPs	
VTN	Allen AI	2D ViT Spatial + Longformer Temporal	~11M	~50 GFLOPs	78.8%
TimeSformer-S	Meta / FAIR	Divided Space/Time Attention	25M	~60 GFLOPs	~78.5%

Key Architectural Approaches

1. Decoupled Attention (TimeSformer, ViViT, VTN)

Separates the mathematics entirely. The model computes spatial attention on individual frames first, and handles temporal attention across the frames sequentially, circumventing multi-dimensional matrix scaling limits.

2. Localized Attention (Video Swin-T, XViT)

Restricts the self-attention mechanism to a small local temporal or spatial window rather than forcing every token to compute relationships with every other token globally, resulting in strictly linear memory scaling.

3. Token Reduction (VideoMAE, TokenLearner, SVT)

Aggressively drops tokens. Whether through massive 90% space-time masking (VideoMAE) or learning to dynamically discard static background pixels (TokenLearner), these architectures simply process vastly less data per frame.

4. Frozen Backbones (EVL)

Locks a heavy image transformer so it requires zero gradients during training, attaching a tiny, trainable temporal decoder on top to learn motion dynamics. Highly efficient for rapid fine-tuning on single-GPU hardware.